

GigaDevice Semiconductor Inc.

GD32A712AEVB

Arm[®] Cortex[®]-M7 32-bit MCU

User Guide

Revision 0.0

(Jul. 2025)

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1. Summary

GD32A712AEVB uses GD32A712AIT3 as the main controller. It uses GD-Link Type-C interface or DC-12V connector to supply power. Reset, User KEY, LED, ADC, DAC, CAN, LIN, I2S, I2C, QSPI, SPI and UART to RS232 interface are also included. For more detail please refer to GD32A712AEVB-LQFP176-V1.1 schematic.

2. Function Pin Assign

Table 2-1. Function pin assignment

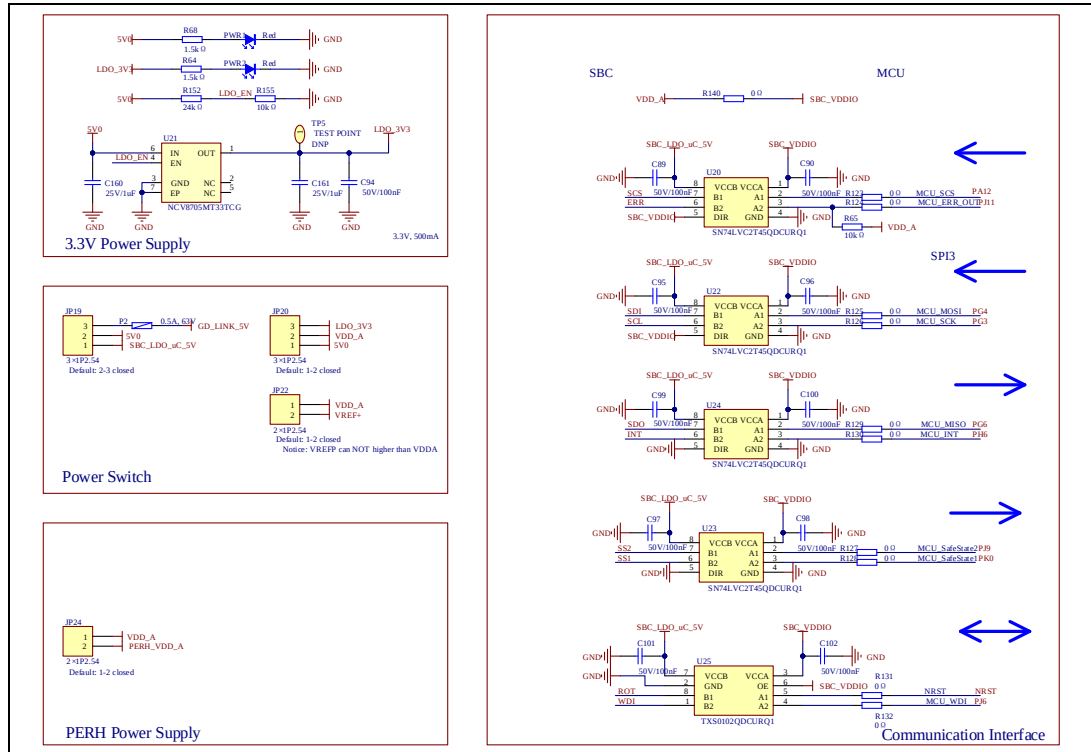
Function	Pin	Description
LED	PE13	LED1
	PE12	LED2
	PE11	LED3
	PE10	LED4
RESET		RESET
USERKEY	PC13	KEY1
	PL3	KEY2
	PL4	KEY3
	PL5	KEY4
ADC	PD11	ADC012_IN9
DAC	PI9	DAC_OUT0
CAN	PH12	CAN0_RX
	PH10	CAN0_TX
	PM3	CAN2RX
	PM2	CAN2TX
	PB12	CAN4RX
	PH11	CAN4TX
	PE14	CAN5RX
	PE15	CAN5TX
LIN	PG0	LIN0_RX
	PF14	LIN0_TX
	PE8	LIN1_RX
	PE7	LIN1_TX
	PD14	LIN2_RX
	PD13	LIN2_TX
	PF13	LIN3_RX
	PJ4	LIN3_TX
I2C	PB2	I2C1_SCL
	PB0	I2C1_SDA
I2S	PB4	I2S_SD

Function	Pin	Description
	PM4	I2S_CK
	PM5	I2S_WS
	PH9	I2S_MCK
QSPI	PH3	QSPI_DQS
	PA4	QSPI_CLK
	PA3	QSPI_IO0
	PH2	QSPI_IO1
	PH5	QSPI_IO2
	PA2	QSPI_IO3
	PA1	QSPI_NSS
	PH4	QSPI_NRST
SPI	PE6	SPI_SCK
	PH13	SPI_NSS0
	PH14	SPI_MOSI
	PE5	SPI_MISO
	PA8	SPI_IO2
	PM0	SPI_IO3
UART (Use LINFlexD UART mode)	PH7	LIN1_TX
	PH8	LIN1_RX
	PJ8	LIN3_TX
	PJ7	LIN3_RX

3. Hardware layout overview

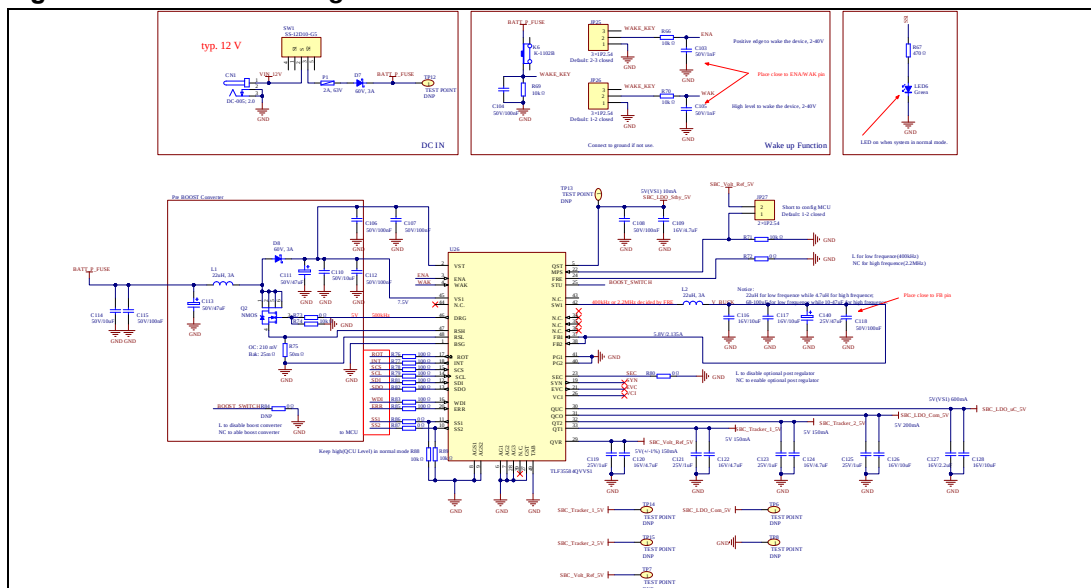
3.1. Power supply

Figure 3-1. Schematic diagram of power supply



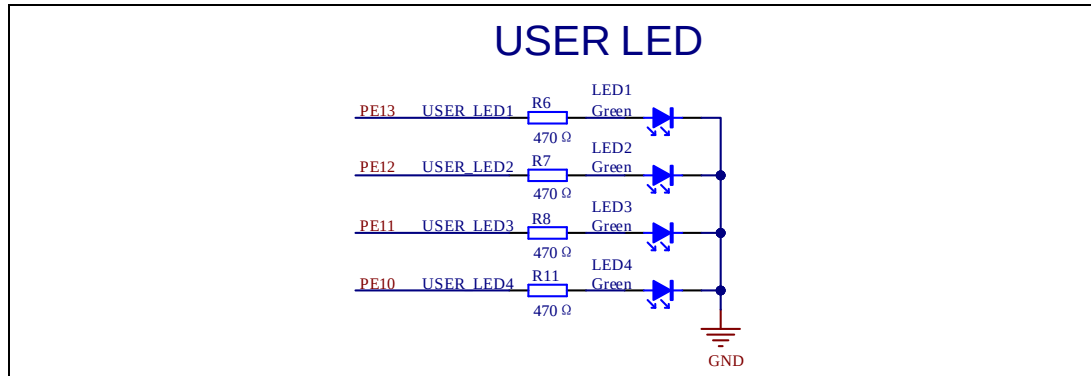
3.2. SBC

Figure 3-2. Schematic diagram of SBC



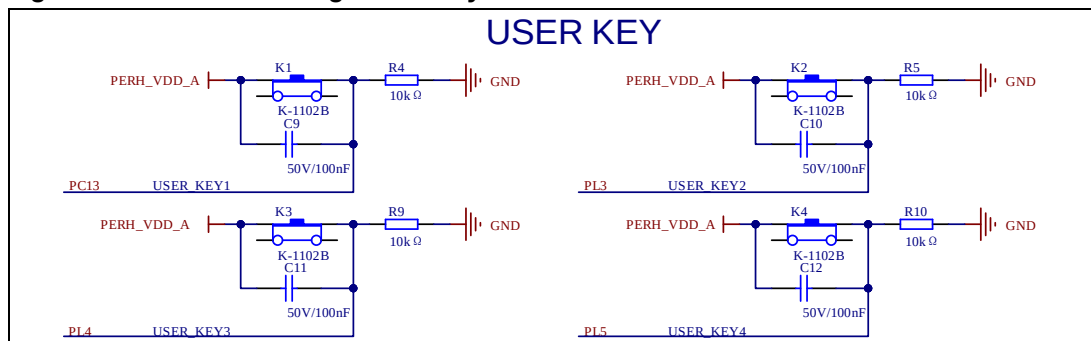
3.3. LED

Figure 3-3. Schematic diagram of LED function



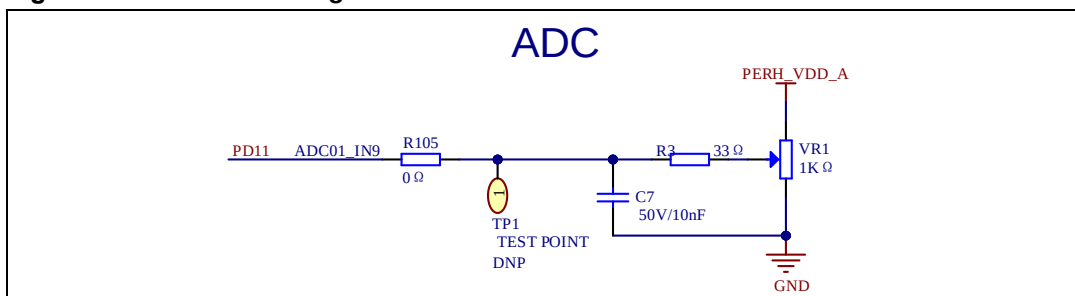
3.4. KEY

Figure 3-4. Schematic diagram of Key function



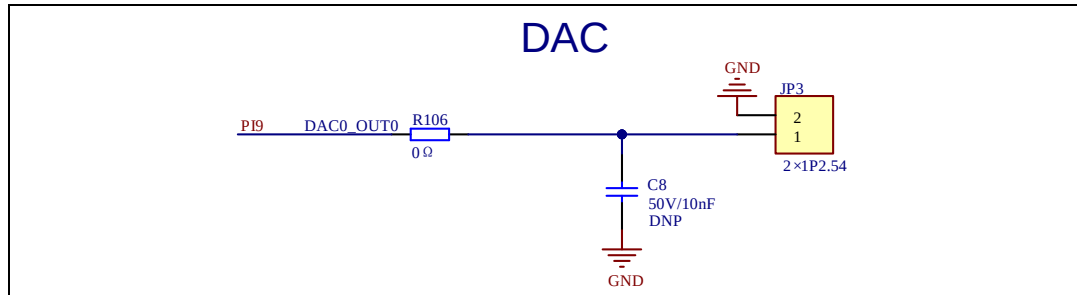
3.5. ADC

Figure 3-5. Schematic diagram of ADC



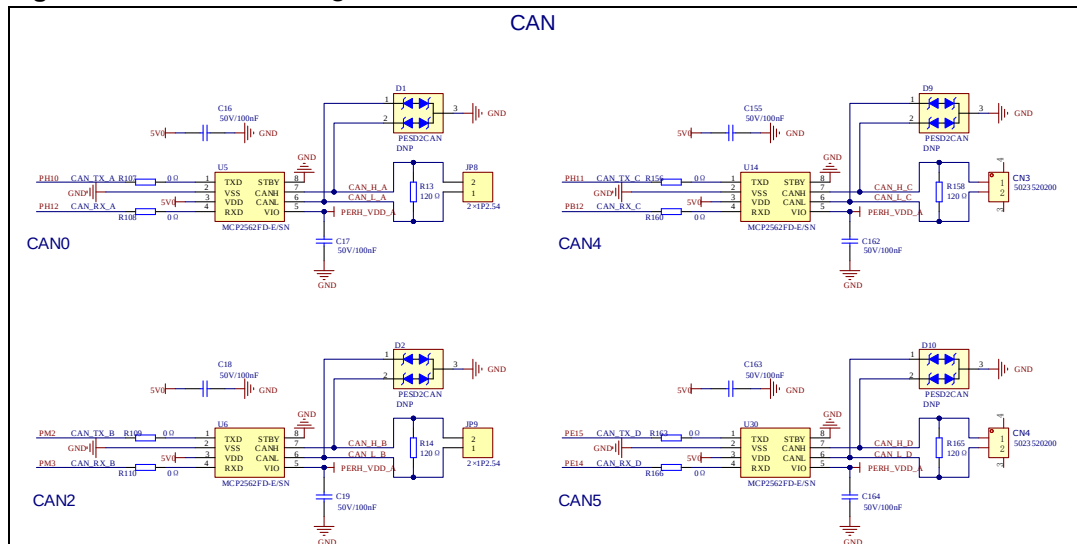
3.6. DAC

Figure 3-6. Schematic diagram of DAC



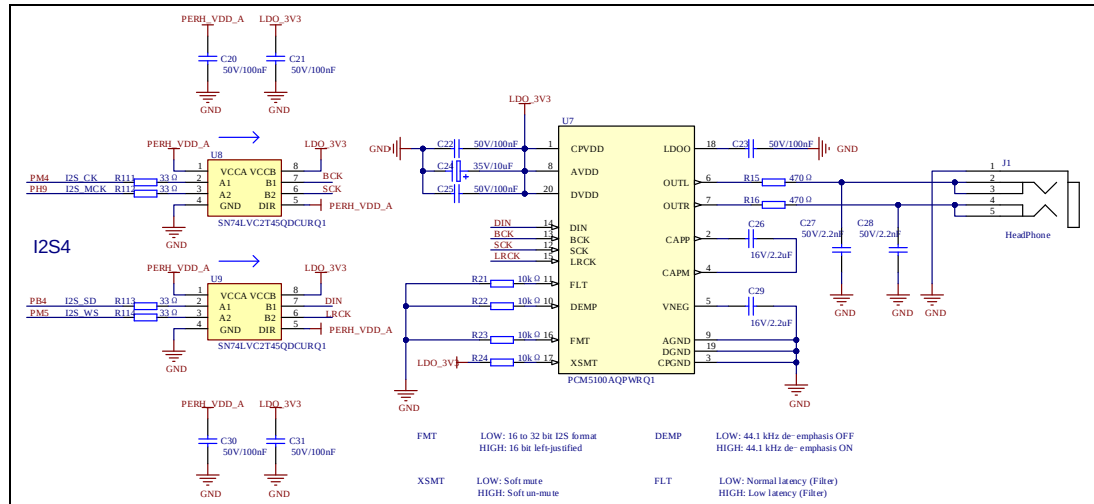
3.7. CAN

Figure 3-7. Schematic diagram of CAN



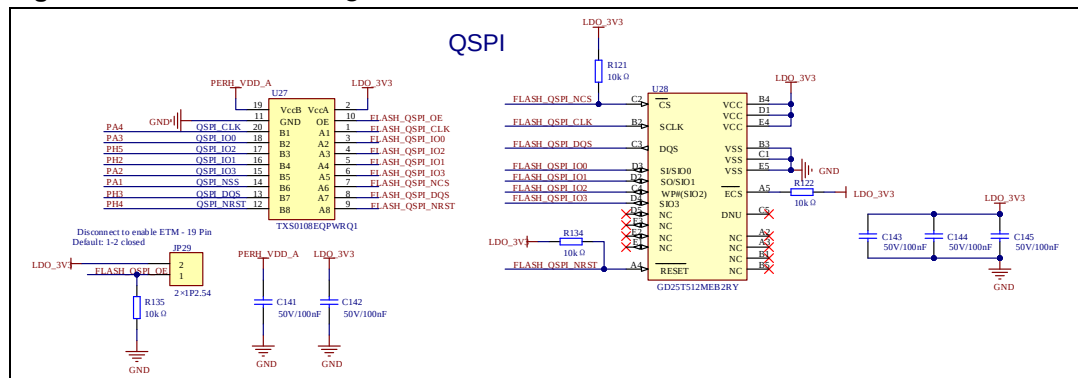
3.10. I2S

Figure 3-10. Schematic diagram of I2S



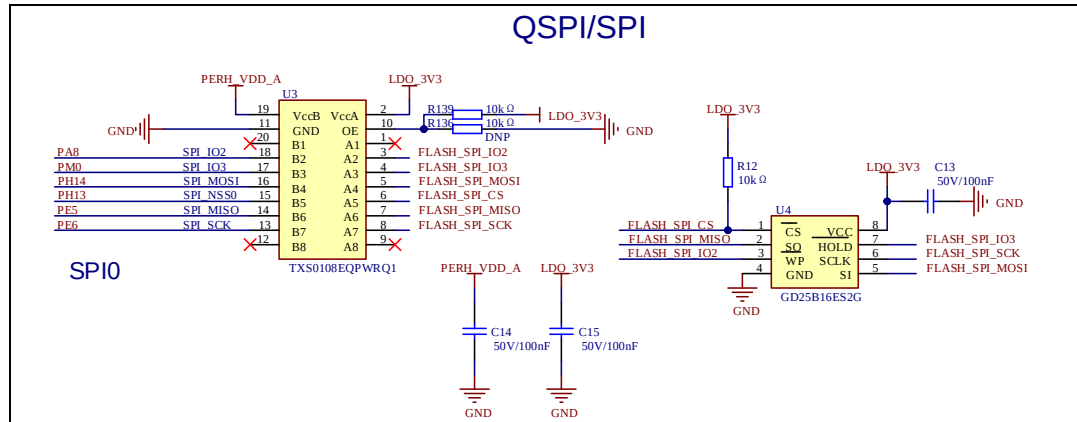
3.11. QSPI

Figure 3-11. Schematic diagram of QSPI



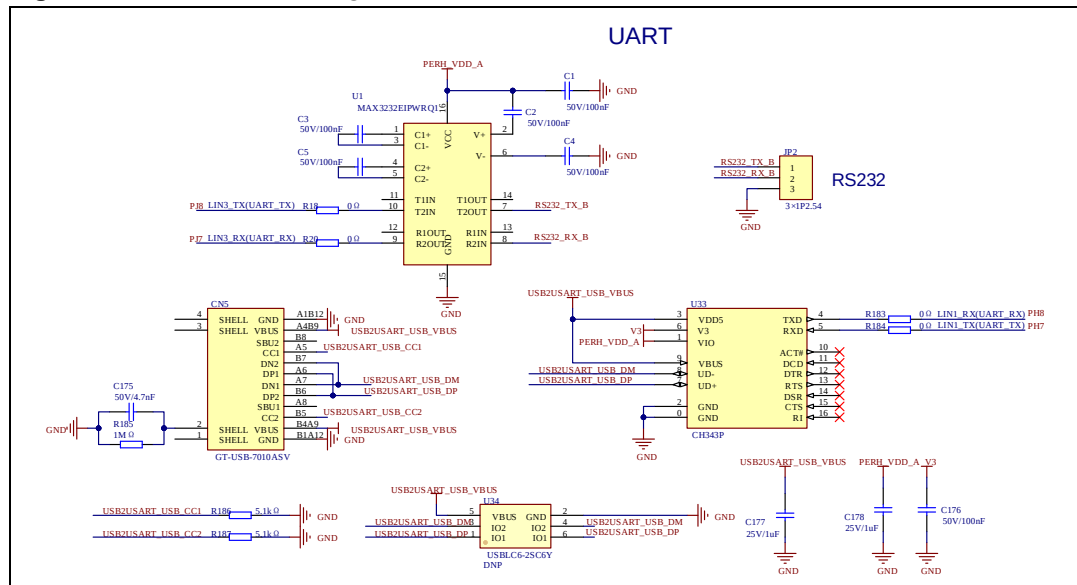
3.12. SPI flash

Figure 3-12. Schematic diagram of SPI flash



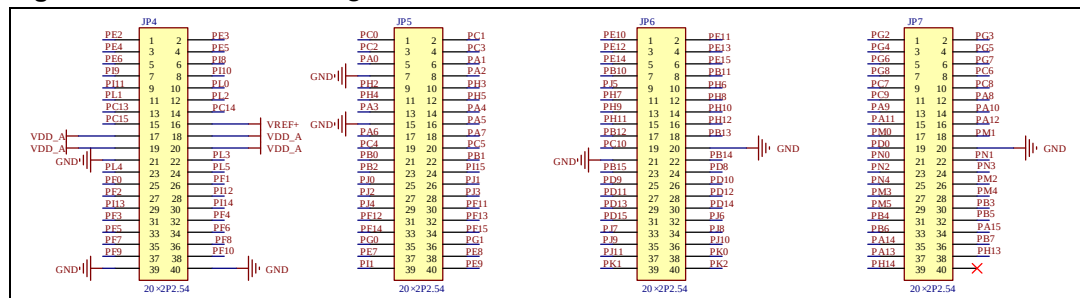
3.13. LINFlexD UART

Figure 3-13. Schematic diagram of UART



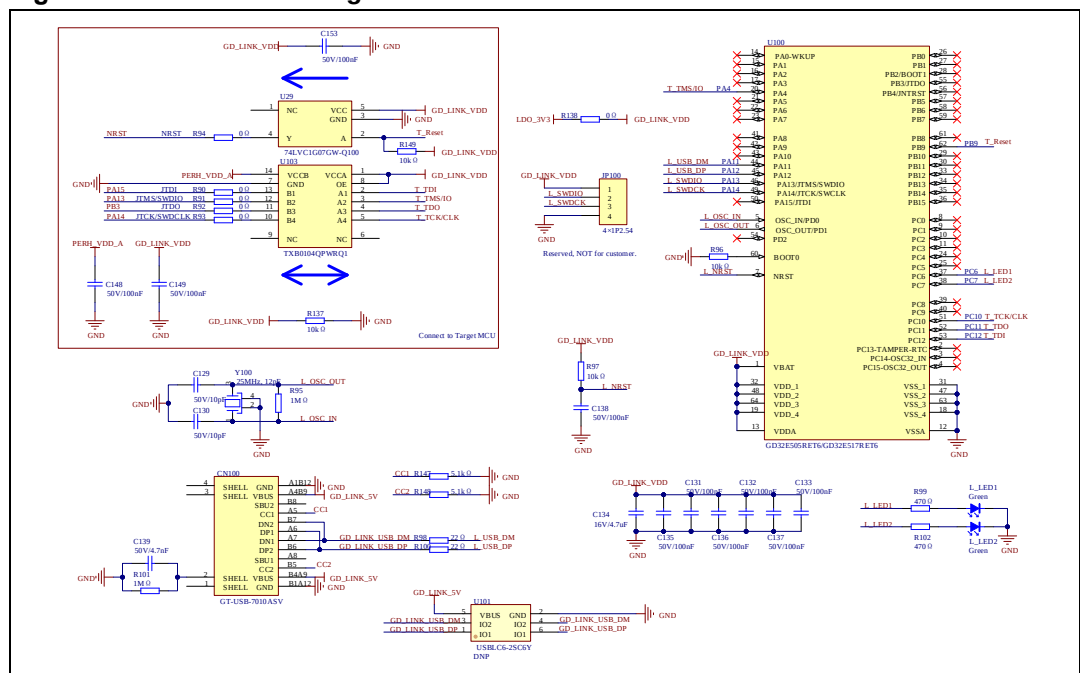
3.14. Extension

Figure 3-14. Schematic diagram of Extension



3.15. GD-Link

Figure 3-15. Schematic diagram of GD-Link



4. Revision history

Table 4-1. Revision history

Revision No.	Description	Date
0.0	Initial Release	Jul.20, 2025

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